

Client: Chainmail Solutions Inc. **Target Partner:** Shenzhen Fine Offset Electronics Co., Ltd. **Date:** June 7, 2026 **Document Ref:** CM-HW-SPEC-2026-V1.0

1. Project Objective & Scope

Chainmail Solutions is launching a decentralized parametric insurance network. We are seeking an established OEM/ODM partner to manufacture, white-label, and factory-configure our weather sensor suite, branded as the **Chainmail Sentinel Series**.

The hardware must support direct cloud telemetry reporting, automated onboarding via factory-mapped serial numbers, and customized industrial casings.

2. Hardware Specifications & SKUs

We require white-labeled equivalents of the following standard Fine Offset / Ecowitt models:

SKU 1: Sentinel Gateway Hub (Model S-100)

Based on Ecowitt GW1100/GW1200

- **Core Function:** Receives RF telemetry from outdoor sensors and transmits normalized data over Wi-Fi (802.11 b/g/n) to the Chainmail ingestion API.
- **Internal Sensors:** Barometric Pressure, Temperature, Humidity.
- **Frequencies:** 915 MHz (US Market) and 868 MHz (EU Market).
- **Power:** 5V USB (Type-C preferred).

SKU 2: Sentinel CAT-7 Outdoor Sensor Array (Model S-700)

Based on Ecowitt Wittboy (WS90)

- **Core Function:** Solar-powered 7-in-1 outdoor weather station with built-in supercapacitor.
- **Sensors:** Ultrasonic Wind Speed, Wind Direction, Rain (piezoelectric rain sensor), Temperature, Humidity, UV, and Light.

- **Frequency Compatibility:** Paired with Model S-100 (915 MHz / 868 MHz).
- **Mounting:** Standard bracket for pole mount.

SKU 3: Sentinel Soil Saturation Probe (Model S-WH51)

Based on Ecowitt WH51

- **Core Function:** Wireless capacitive soil moisture sensor.
- **Frequency Compatibility:** 915 MHz / 868 MHz.

3. Firmware Customization & Telemetry Protocol

To support zero-configuration onboarding, gateways must be pre-flashed with custom default firmware settings at the factory.

Custom Server Configurations

The gateway firmware must be set with the following default values in the **Custom Server** settings:

Parameter	Configuration Value
Protocol Type	Ecowitt (HTTP POST, URL-encoded)
Server/IP	api.chainmailsolutions.ai
Path	/api/ingest/ecowitt
Port	443 (HTTPS / SSL Enabled)
Upload Interval	60 seconds
Default Password/Secret	Pre-flashed unique 32-character HMAC token mapped to MAC address

IMPORTANT

HTTPS Support:

Gateways must support secure TLS 1.2 or TLS 1.3 handshakes when posting data to our cloud API endpoint.

CHAINMAIL SENTINEL SYSTEM

3. Scan QR code below to bind device to your profile.

[QR CODE]

MAC Address: AA:BB:CC:11:22:33

Node Activation Key: [32-Character Unique Token]

+-----+

Factory Logging Requirement

For every unit shipped, the manufacturer must provide a secure CSV inventory file containing:

1. Device MAC Address
2. Device Serial Number
3. Generated Activation Token
4. Printed QR Code Payload URL (e.g., <https://app.chainmailsolutions.ai/activate?mac=AABBCC112233&token=...>)

6. Business Terms & RFQ Questions

Please provide pricing and lead times based on the following volumes:

- **Initial Minimum Order Quantity (MOQ):** 1,000 units (Combo Kits containing 1x S-100 and 1x S-700).
- **Secondary Volume Pricing:** 5,000 units and 10,000 units.
- **Sample Units:** Lead time and cost for 5 custom-branded prototype samples (915 MHz).
- **Certification:** Confirmation of FCC (Part 15) and CE compliance for all RF transmitters.

RFQ Submission Instructions

Please submit technical specifications, material declarations, and unit price quotes in USD (FOB Shenzhen) to:

Greg Narcisi

Co-Founder, Chainmail Solutions Inc.

greg.narcisi@chainmailsolutions.ai